What do zoning and adopted plans add to predicting where cities build?

Evidence from official parcel data in seven US cities

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Abstract

Planning support tools project land-use scenarios by combining what exists, what zoning permits and what the plan intends, yet what each layer contributes is rarely tested. We assembled the official existing land use, adopted plan and zoning for 4,995,208 parcels in ten large US cities, including a plan published only as PDF maps (digitised with median errors of 1.1–7.2 ft). In seven cities with parcel histories, models fitted on one period were scored on the next, adding one layer at a time. Existing conditions ranked the lots that developed with an AUC of 0.824–0.911 in six cities (0.571 in the seventh). Context added most: +0.020 to +0.116 where significant (five cities). Over context, zoning and the plan added at most +0.021, except in Phoenix (+0.049 and +0.054), and two of these gains disappeared when lots rezoned during the study were excluded. Zoning and plan classes predicted what developed lots became better than their existing use. Capacity build-out exceeded the homes built 76- to 193-fold; likelihood weighting gave 0.81–2.64 times the observed total; a five-year trend ranked districts better. Regulatory layers inform what development becomes and how much it could hold; existing conditions and context predict where.

Keywords: planning support systems; zoning; land-use plans; urban development prediction; digital twins; model validation

1 Introduction

Land-use decisions in US cities rest on projections. Rezonings, amendments to comprehensive plans and housing targets are assessed by estimating the development they will bring. Planning support systems and scenario tools make these estimates by overlaying parcel layers (existing land use, zoning and the adopted plan) and, increasingly, a model of what is likely to happen (Klosterman, 1999; Waddell, 2002; Geertman and Stillwell, 2020). The common default is capacity: each eligible site is assumed to build out to what the rules allow. Capacity answers a legal question, what a regulation permits, transparently. It is also used to answer a different question: what will be built.

There are reasons to expect the two answers to diverge. Development is rare and selective: in the one- and two-year windows studied here, 0.06% to 1.3% of lots developed. Causal studies disagree on whether upzoning changes construction in the short run. In Chicago, upzoned parcels rose in value without more permitted homes within five years (Freemark, 2020), whereas upzoning raised construction in Portland and Auckland (Dong, 2024; Greenaway-McGrevy and Phillips, 2023). Evaluations of plan implementation find that adopted plans are followed only in part (Talen, 1996; Laurian et al., 2004; Brody and Highfield, 2005; Berke et al., 2006). In the three Manhattan rezonings replayed below, a capacity screen that reproduces the environmental reviews' own projections on their projected sites (0.86 and 1.00 times the review figures) expected 8,423 homes on development sites. By 2025, 561 had been built.

The predictive value of each regulatory layer is nonetheless seldom measured. Land-change models now incorporate zoning and planning regulations (Domingo et al., 2021; Abolhasani et al., 2026), but they use them to simulate scenarios, not to test what the regulations add on held-out history. Reviews of planning support systems and urban digital twins find that claimed capabilities run ahead of demonstrated ones (Vonk et al., 2005; Azadi et al., 2025; Kruse et al., 2026). The question matters for practice because zoning and plans are the layers that planners set. If they add little to predicting where development happens, scenario tools built on them will misjudge where change occurs and how much of it a decision will bring. If they add much, tools should give them weight.

A practical barrier is data. The three layers are published by different agencies, in different schemas, and some plans exist only as documents: Philadelphia's land-use plan exists only as maps in 17 PDF district plans. A test across cities therefore first requires the layers on one schema, with every translation reported.

This paper makes four contributions:

1. **Data.** The official existing land use, adopted plan and zoning for 4,995,208 parcels in ten of the largest US cities, on one 16-class schema, with ingestion checks: homes are within 0.95–1.06 times the Census 2020 count in every city. It also includes a validated method to digitise plans published as vector PDF maps.

2. **Test.** A cross-city, out-of-time component ablation of what zoning and the adopted plan add to existing conditions and neighbourhood context, in seven cities and 3,346,335 lots. It covers three planning questions: where development happens, what it becomes and how many homes it adds. It controls for layers that postdate the outcomes.

3. **Scenario evidence.** Capacity build-out, likelihood-weighted build-out and a simple trend, compared with what was built in four cities and in three Manhattan rezonings.

4. **Open implementation.** PlanArena, a parcel-level planning support system that generates every number reported here from public sources and exports layers and scenarios to ArcGIS Urban.

The results are predictive. They show how much each layer helps to anticipate development, not what zoning or plans cause. Section 2 reviews related work, Section 3 the data, Section 4 the methods and Section 5 the results; Section 6 discusses implications and limits.

2 Related work

Predicting urban development. Spatial models of urban growth range from cellular automata calibrated on historical urbanisation (Clarke et al., 1997) to site-level statistical models of land-use change (Landis and Zhang, 1998) and behavioural simulations of households, firms and developers interacting through land markets (Waddell, 2002; Waddell et al., 2003). Agent-based models represent the decisions behind land change (Parker et al., 2003). Reviews list neighbourhood effects, temporal dynamics and cross-scale interactions as priorities (Verburg et al., 2004; Brown et al., 2013; Weerasinghe et al., 2025). At the parcel level, autologistic models capture existing and historical neighbourhood conditions (Tepe and Guldmann, 2020), and machine learning and econometric models of redevelopment are complements rather than substitutes (Carruthers and Wei, 2024). Urban economics ties redevelopment to the value of land relative to that of the existing structure (Rosenthal and Helsley, 1994; Munneke, 1996; Dye and McMillen, 2007). Parcel studies in Chicago find that smaller and older buildings with less lot coverage are demolished and replaced more often (Weber et al., 2006; Charles, 2013). These findings motivate the existing-conditions and context features used here.

Regulations and plans as predictors. Land-use regulation affects new construction (Mayer and Somerville, 2000), but rezoning is not random: in Los Angeles, upzoning occurred where development opportunity met weak political resistance (Gabbe, 2018). Current zoning may therefore reflect development that was already coming, which complicates any test of zoning against past development. Studies of upzoning estimate causal effects in one city at a time and differ in their conclusions (Freemark, 2020; Dong, 2024; Greenaway-McGrevy and Phillips, 2023). Plan evaluation distinguishes conformance, whether outcomes match the plan, from performance, whether the plan shaped decisions (Talen, 1996; Laurian et al., 2004; Berke et al., 2006; Loh, 2011; Feitelson et al., 2017). Land-change science has called for planning to be represented explicitly in models (Hersperger et al., 2018). Two recent studies do so:

- Domingo et al. (2021) use the floor area each zoning plan allows per parcel in multi-scenario simulations of four Spanish urban areas;
- Abolhasani et al. (2026) standardise local planning regulations for 3D land-use simulation in Zurich and Copenhagen.

Both integrate regulations into simulation; neither measures what the regulations add to prediction on held-out history.

Planning support systems and urban digital twins. Planning support systems combine land suitability, demand projection and allocation in scenario planning (Klosterman, 1999; Geertman and Stillwell, 2020). Their limited use and uncertain usefulness are long-standing concerns (Vonk et al., 2005; te Brömmelstroet, 2013; Pelzer, 2017). Urban digital twins extend the idea to integrated city models (Batty, 2018, 2024; White et al., 2021; Lehtola et al., 2022; Ferré-Bigorra et al., 2022). Reviews report technical and organisational barriers (Lei et al., 2023; Wu and Guan, 2025) and a gap between the ambitions of urban twins and their demonstrated contributions to planning (Azadi et al., 2025; Luo et al., 2025; Kruse et al., 2026). Optimisation and allocation methods generate and rank land-use alternatives (Malczewski, 2006; Rahman and Szabó, 2021; Zheng et al., 2023; Feng et al., 2024; Lin et al., 2025). This paper asks a prior question: what the regulatory layers on which such methods operate add as evidence.

Validation. Land-change models should be compared with a null model of persistence, at several resolutions (Pontius et al., 2008), with disagreement split into quantity and allocation (Pontius and Millones, 2011). Their transition probabilities should be validated directly, not only through simulated maps (Black et al., 2023). Under spatial and temporal dependence, random cross-validation overstates skill (Roberts et al., 2017; Ploton et al., 2020), and a model applies only where new data resemble its training data (Meyer and Pebesma, 2021). We therefore test out of time and across cities (Wei et al., 2016), and resample spatial tiles to estimate uncertainty (Kunsch, 1989).

Map digitisation. Digital map processing and automatic georeferencing have been developed mainly for scanned raster maps (Chiang et al., 2014; Luft and Schiewe, 2021; Milleville et al., 2022). Plans exported from GIS as vector PDFs keep each lot as a filled shape, so the shapes can be read directly and registered to parcels with standard image and point registration (Reddy and Chatterji, 1996; Besl and McKay, 1992; Umeyama, 1991).

Table 1: The ten cities: official parcel layers on one schema. Shares are of the city's parcels.

City	Parcels	Existing use (source)	Share	Adopted plan	Share	Zoning	Homes / Census
New York	856,634	MapPLUTO (City Planning)	100%	none citywide	–	100%	1.03
Los Angeles	672,444	SCAG 2020; county assessor	100%	General Plan (SCAG 2019)	100%	100%	0.99
Chicago	606,894	CMAP 2023; county assessor	100%	none citywide	–	96%	1.05
Houston	630,525	City (appraisal codes)	97%	none	–	–	0.95
Phoenix	510,871	County assessor; MAG	100%	General Plan	100%	99%	0.99
Philadelphia	547,410	City Planning Commission; OPA	100%	district plans (digitised PDF)	83%	100%	0.99
San Antonio	438,009	Bexar Appraisal District	100%	adopted area plans	61%	82%	0.99
San Diego	272,948	SANDAG 2025; county assessor	100%	General and community plans	100%	94%	1.02
Dallas	311,431	City 2026; DCAD	100%	forwardDallas! 2.0	99%	79%	0.98
Milwaukee	148,042	MPROP	100%	PDF only (not included)	–	99%	1.06
All	4,995,208						

Existing use is the city's or regional agency's parcel land use, refined by assessor records where the layer groups uses. Homes / Census: homes in the twin over Census 2020 housing units. Houston has no zoning. Milwaukee's area plans are published as PDFs that could not be downloaded by script.

3 Data

3.1 Ten cities, three official layers

The study covers the nine largest US cities other than San Jose, and Milwaukee (Table 1). For each city, the existing land use is the parcel land use published by the city or its regional planning agency. Where that layer groups uses, it is refined with assessor records: unit counts split single-family from multi-family housing, and building classes identify offices, hotels, schools and warehouses. The adopted plan (future land use) and zoning are separate layers from the same or another official source. Every source code is translated by an explicit crosswalk into 16 classes, which nest in nine analysis groups:

- housing: single-family, two to four homes, five or more;
- mixed use, retail, office, hotel;
- industry and warehousing;
- education, health and civic uses;
- open space, transportation, parking and vacant land.

Floor area, homes, year built and assessed value come from assessor or city records. Where records lack floor area, it is estimated from building footprints and heights and flagged. Jobs are fitted to the Census Bureau's LEHD block counts. Every input is downloaded by script, with its URL and SHA-256 hash recorded.

Existing use covers at least 97% of parcels in every city. Zoning is available in nine cities (Houston has none) and an adopted plan in six. New York, Chicago and Houston publish no citywide land-use map, and Milwaukee's area plans are PDF documents that could not be retrieved by script. Where both layers exist, the existing use already lies in the plan's analysis group on 83%–92% of parcels. Plans mostly confirm what exists; their information lies in the minority of parcels where they differ.

3.2 Digitising a plan published as PDF maps

Philadelphia2035's land-use recommendations exist only as "Proposed Land Use" maps in its district plans: 17 documents covering 18 districts. The maps are vector drawings in which each lot is a filled shape in one of the City Planning Commission's categories. We convert them in six steps (Figure 1):

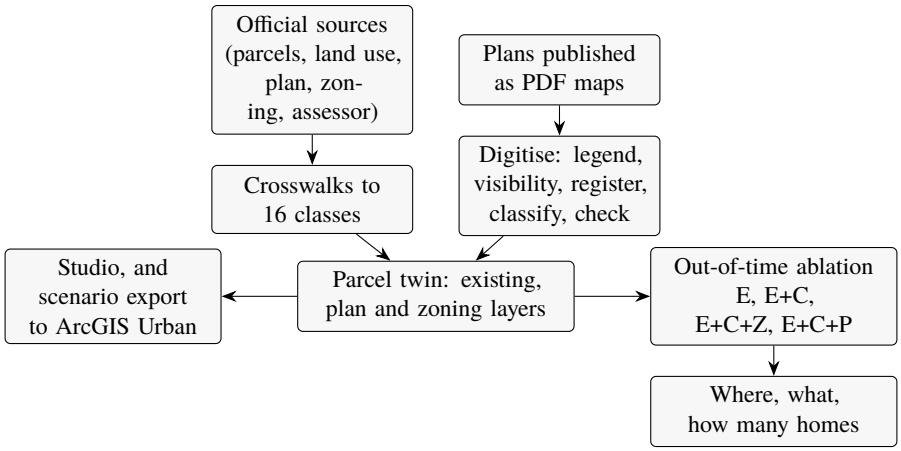


Figure 1: Data and test pipeline. Every source and translation is recorded; nothing in the three official layers is predicted.

1. **Legend.** The legend is read: each swatch's fill colour, with the label to its right.
2. **Colours.** Map colours are matched one to one to legend colours by perceptual distance (CIELAB ΔE, Hungarian assignment), because a legend may be CMYK while the map is RGB. Pairs further apart than $\Delta E = 45$ are not matched.
3. **Visibility.** Maps often draw a wider area and clip it to the district, and the vector shapes carry no clipping. Each shape is kept only if the rendered page shows its colour at a point inside it, and only the largest map frame on the page spread is kept, which removes insets.
4. **Registration.** A similarity transform (scale, rotation, translation) is fitted. The rendered map is first matched as an image to the footprint of the district's parcels and to the district outline, searching rotation and scale, finding translation by FFT and scoring by Dice overlap. The fit is then refined by iterative closest point from each parcel centroid to the nearest shape centroid.
5. **Classification.** Each parcel takes the category of the smallest shape containing its interior point.
6. **Checks.** A map is used only if (a) the median parcel-to-shape distance is at most 10 ft and (b) its plan agrees with the existing use clearly more often than the same map shifted 300 ft. Because most land keeps its use in a plan, a misregistered map fails this falsification test.

Maps from 15 of the 17 plans passed, covering 16 of 18 districts and 83% of the city's parcels. Median registration errors were 1.1–7.2 ft (median 3.2 ft), less than the width of a typical rowhouse lot. The maps covered 88%–100% of each district's parcels. The digitised plans agree with the existing use on 60%–90% of parcels, against 42%–66% for the shifted maps, a margin of at least 17 points in every district. The maps of South and West Park also draw neighbouring districts, which defeated the registration; those districts have no plan class. Philadelphia has no parcel history in this study, so its plan layer serves the system but not the test below.

3.3 Parcel histories

Seven cities have parcel histories that allow an out-of-time test (Table 2). The definition of development follows what each source records:

Table 2: Test panels. Models are fitted on the training window and scored on the test window.

City	Development	Source	Train	Test	Lots	Developed	Rate
Los Angeles	new building or floor area +25%	county assessor rolls	2021–2023	2023–2025	667,844	2,222	0.33%
Dallas	new building or floor area +25%	appraisal district files	2022–2024	2024–2026	291,841	3,916	1.34%
New York	new building or floor area +25%	MapPLUTO	2021–2023	2023–2025	856,901	4,064	0.47%
Milwaukee	new building	MPROP	2021–2023	2023–2025	147,539	85	0.06%
Phoenix	change of use group	MAG land-use inventory	2022–2024	2024–2026	503,635	1,835	0.36%
San Diego	change of use group	SANDAG land-use inventory	2023–2024	2024–2025	272,281	202	0.07%
Chicago	new-construction permit	building permits	2021–2023	2023–2025	606,294	1,548	0.26%

Lots already under construction at the start of a window are left out. Floor area +25% also requires +2,000 sq ft. Change of use group: the lot changes to a built use of another of the nine groups, or gains homes.

- a new building, or floor area up by more than 25% and 2,000 sq ft, in Los Angeles (county assessor rolls, 2021–2025), Dallas (appraisal district files, 2022–2026) and New York (MapPLUTO, 2021–2025);
- a new building in Milwaukee, where recorded floor area is not consistent across years;
- a change to a built use of another group, or more homes, between the regional agencies' land-use inventories in Phoenix and San Diego;
- a new-construction permit issued in Chicago.

Lots already under construction at the start of a window are excluded, and every feature is measured at the start of its window. The test windows contain 3,346,335 lots, of which 13,872 developed.

4 Methods

Feature blocks. Four blocks are defined identically in every city, from portable classes and ratios without city-specific codes:

- **E, existing conditions:** the analysis group of the existing use, lot area, floor area, built floor-area ratio, building age, homes and the land share of assessed value;
- **C, context:** the share of lots within 400 m that were built or changed use in the five years before the window, the number of lots within 400 m and their land-use mix;
- **Z, zoning:** the zoning class, whether the existing use departs from it, and the step in housing intensity from the existing use to the zoning class;
- **P, adopted plan:** the same three features for the plan class.

Out-of-time ablation. For each city and block set, a gradient-boosted classifier (LightGBM; 400 trees, learning rate 0.04, 31 leaves) is fitted on the training window and scores the lots of the test window. The block sets are E, E+C, E+C+Z, E+C+P and E+C+Z+P, together with the sets without context. We report the area under the ROC curve (AUC), with average precision and precision among the top 1% of lots in Appendix Table 9. The gain from adding a block is the difference in AUC; its 95% interval comes from 200 bootstrap resamples of 2 km tiles, which preserves spatial dependence. As a second view, all sets are also cross-fitted over five folds of 2 km tiles within the full period.

Leakage control. In Los Angeles both layers were compiled in 2019, before every window. New York and Milwaukee have zoning from each year's records. In Phoenix, San Diego and Chicago the zoning or plan layer is current, but zoning cases and plan polygons are dated. A conservative variant therefore excludes, in each window, lots whose zoning or plan dates from the window's start or later. Dallas's current layers carry no dates, so its zoning and plan gains are upper bounds.

Destination. Among lots that developed in the test window, the class they became is predicted by the existing use (no change), the zoning class, the plan class and models cross-fitted over tiles. Each is scored as one of 16 classes and as one of nine groups.

Homes. In the four cities with unit counts at both ends of the window, homes added by district are compared with four projections:

- **zoning build-out**, in which every lot reaches the homes per acre that the 90th percentile of lots of its zoning class already reaches in the city;
- **plan build-out**, the same for the plan class;
- **likelihood-weighted build-out**, the same increments multiplied by each lot's out-of-time probability of development;
- **the trend**, homes completed in the five years before the window, scaled to its length.

We report the ratio of projected to observed total and the Spearman correlation across districts.

Transfer. For the five cities whose outcome is a building or a permit, a model fitted on the other cities' training windows scores the city's test window and is compared with the city's own model. A missing plan layer is coded as unknown.

Manhattan rezonings. To test the approach at the scale of a single decision, we replay three Manhattan rezonings: East Harlem (2017), Inwood (2018) and SoHo/NoHo (2021).

- **Sites** are lots that can add floor area under the adopted zoning.
- **Capacity** assumes full build-out.
- **The likelihood-weighted projection** multiplies each site's homes by its probability of redevelopment by 2025. The probability comes from a redevelopment model trained only on outcomes known by the adoption year, scored with the adopted zoning in place.

That model was tested separately on Manhattan's annual records. Across 15 rolling origins (2016–2025, with horizons of one, three and five years), its out-of-time AUC was 0.847–0.952, above the best simple baseline at every origin.

Table 3: Where development happens: out-of-time AUC by block set. E existing conditions, C neighbourhood context, Z zoning, P adopted plan; – where the city has no such layer.

City	E	E+C	E+C+Z	E+C+P	E+C+Z+P	Best set (any)
Los Angeles	0.864	0.892	0.894	0.892	0.894	E+C+Z+P
Dallas	0.898	0.918	0.922	0.920	0.923	E+C+Z+P
New York	0.892	0.914	0.919	–	–	E+C+Z
Milwaukee	0.911	0.896	0.924	–	–	E+Z
Phoenix	0.859	0.793	0.842	0.847	0.853	E+Z+P
San Diego	0.571	0.687	0.684	0.707	0.703	E+C+P
Chicago	0.824	0.888	0.891	–	–	E+C+Z

Best set: the highest AUC among all fitted combinations, including those without context.

Table 4: Gain in out-of-time AUC from adding a block, with 95% bootstrap intervals over 2 km tiles. Bold: the interval excludes zero.

City	C over E	Z over E+C	P over E+C	Z, conservative	P, conservative
Los Angeles	**+0.028 [0.020, 0.036]**	+0.002 [−0.001, 0.005]	0.000 [−0.004, 0.003]		
Dallas	**+0.020 [0.015, 0.025]**	**+0.004 [0.002, 0.005]**	**+0.002 [0.001, 0.003]**		
New York	**+0.023 [0.018, 0.029]**	**+0.004 [0.003, 0.007]**	–		
Milwaukee	−0.015 [−0.079, 0.048]	+0.029 [−0.031, 0.075]	–		
Phoenix	−0.066 [−0.139, 0.001]	**+0.049 [0.009, 0.091]**	**+0.054 [0.005, 0.100]**	**+0.055 [0.017, 0.088]**	**+0.041 [0.004, 0.073]**
San Diego	**+0.116 [0.053, 0.170]**	−0.003 [−0.022, 0.018]	**+0.021 [0.003, 0.039]**	−0.009 [−0.029, 0.015]	−0.011 [−0.033, 0.007]
Chicago	**+0.063 [0.049, 0.078]**	**+0.004 [0.001, 0.007]**	–	−0.001 [−0.005, 0.003]	–

Conservative: in each window, lots whose zoning or plan dates from the window's start or later are left out; possible where a current layer records those dates (Phoenix, San Diego, Chicago). Dallas's current layers carry no dates, so its zoning and plan gains are upper bounds.

5 Results

5.1 Where development happens

Existing conditions alone rank the lots that developed well (Table 3). Out-of-time AUC with block E is 0.824–0.911 in six cities and 0.571 in San Diego, whose one-year window recorded few changes.

Context adds the most (Table 4, Figure 2). Its gain is significant in five cities (Los Angeles, Dallas, New York, San Diego and Chicago), from +0.020 to +0.116. It is not significant in Milwaukee, and it is negative in Phoenix (−0.066; interval −0.139 to +0.001). Phoenix's regional series starts in 2022, so recent change is unknown in its training window. Its context block therefore reduces to the number and land-use mix of nearby lots, which did not generalise from the training to the test window.

Zoning carries information of its own. Added to existing conditions alone, it raises AUC significantly in five of seven cities (+0.008 to +0.060). Lots that developed were 2.1 to 9.5 times as likely as other lots to have a zoning class outside their existing use group. Most of this information is shared with existing conditions and context, however. Developed lots were also 3.5 to 19 times as likely to be vacant, and once context is included the regulatory layers add little:

- **zoning** adds a significant gain over context in four of seven cities (Dallas, New York, Phoenix and Chicago);
- **the plan** adds one in three of four (Dallas, Phoenix and San Diego);
- outside Phoenix these gains are at most +0.021;

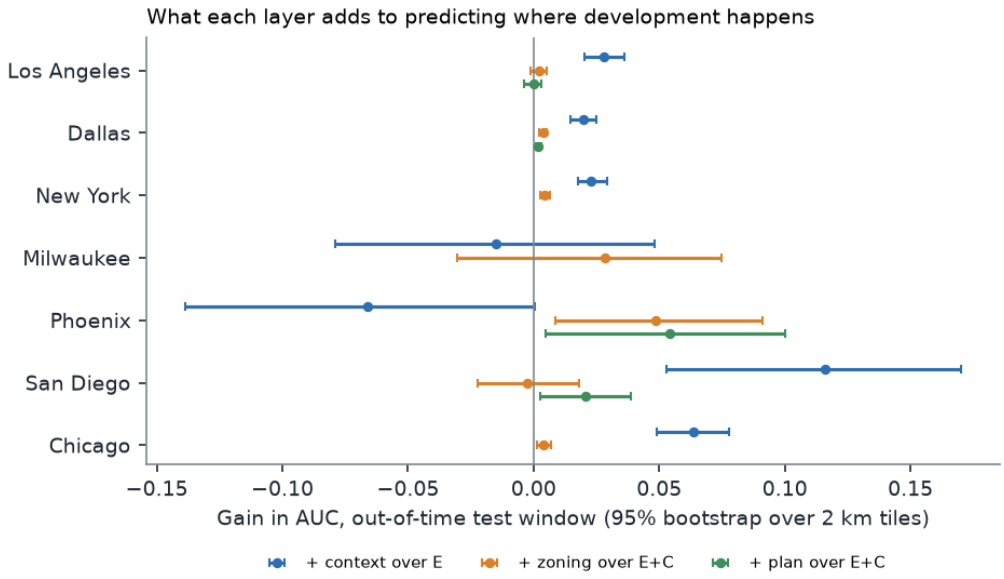


Figure 2: Gain in out-of-time AUC from adding each block, with 95% tile-bootstrap intervals.

- in Los Angeles, where both layers predate the test, zoning and plan together add +0.002, which is not significant.

No layer significantly reduced performance once context was included.

Phoenix is the exception: zoning adds +0.049 and the plan +0.054 over context, and its best block set, E+Z+P (AUC 0.927), omits context. Development there, measured as a change of land-use class in the regional inventory, is concentrated on vacant lots (45% of developed lots against 2% of all lots). It is also concentrated on lots whose zoning or plan class differs from their existing use (66% and 73%, against 7% and 9%).

The conservative variant tests whether such gains come from layers updated during the study. The zoning and plan gains in Phoenix survive it. The plan gain in San Diego and the zoning gain in Chicago do not, so in those cases the apparent value of the layer came from lots whose zoning or plan changed during the study. Spatial cross-fitting gives the same picture: context raises AUC in six of seven cities, and zoning on top of context raises it in six of seven, by small amounts.

5.2 What developed lots become

The regulatory layers matter more for what a developing lot becomes (Table 5). In all six cities with a destination, the zoning class names the new class correctly more often than the existing use does, by 12 to 64 percentage points. (In Phoenix and San Diego, development is a change of class by definition, so there the existing use is never right at class level.) In Dallas, whose plan was adopted in 2024, the plan class is right for 80% of developed lots, against 61% for zoning and 37% for no change. A model combining the layers matches or beats the best single layer in all six cities.

Table 5: What developed lots became: share named correctly, as one of 16 classes / one of nine groups.

City	Developed	No change	Zoning class	Plan class	Model (blocks)
Los Angeles	1,883	57% / 79%	70% / 87%	70% / 87%	74% / 89% (E+C+Z+P)
Dallas	3,910	37% / 40%	61% / 65%	80% / 89%	86% / 95% (E+C+Z+P)
New York	4,002	24% / 37%	49% / 65%	–	54% / 70% (E+Z)
Milwaukee	78	18% / 18%	37% / 77%	–	63% / 76% (E+Z)
Phoenix	1,835	0% / 35%	64% / 74%	47% / 62%	75% / 81% (E+C+Z+P)
San Diego	202	0% / 52%	14% / 74%	27% / 70%	64% / 75% (E+C+Z+P)

In Phoenix and San Diego development is a change of class (to another group, or to more homes), so "no change" is never right at class level there. Models are cross-fitted over 2 km tiles among the developed lots. Chicago's permits do not record the new use.

Table 6: Homes added in the test window: projection over observed total (Spearman correlation across districts).

City	Observed	Zoning build-out	Plan build-out	Likely × zoning	Likely × plan	Five-year trend
Los Angeles	22,853	193× (0.10)	189× (0.24)	0.81× (0.46)	0.81× (0.52)	0.59× (0.70)
Dallas	10,926	141× (0.00)	127× (0.34)	2.64× (0.00)	1.69× (0.30)	1.25× (0.72)
New York	116,776	76× (0.34)	–	0.96× (0.55)	–	0.47× (0.81)
Milwaukee	3,798	115× (−0.23)	–	0.97× (−0.37)	–	0.32× (0.51)

Build-out: every lot at the homes per acre that the 90th percentile of lots of its zoning (or plan) class reaches in the city today. Likely: the build-out increment times the out-of-time probability of development (blocks E+C+Z+P where available). Trend: homes completed in the five years before the window, scaled to its length.

5.3 How many homes

Capacity build-out overstates near-term housing by two orders of magnitude (Table 6). Zoning and plan build-out exceed the homes added in the test window 76- to 193-fold. Part of this gap is definitional, since build-out is an end state rather than a two-year flow, but it shows how far capacity is from what a short horizon delivers. Weighting build-out by the out-of-time likelihood brings the totals to 0.81–2.64 times the observed.

The likelihood-weighted projections rank districts less well than the five-year trend, however. Their Spearman correlations are -0.37 to 0.55, against 0.51 to 0.81 for the trend, which ranks better in all four cities even though its totals are 0.32–1.25 times the observed. Recent local construction is the better guide to which districts will add homes; likelihood-weighted capacity is the better guide to how many.

5.4 Transfer between cities

A model fitted on other cities predicts less well than the city's own model in four of five cities (Table 7). With all blocks, the AUC is lower by 0.046 to 0.070. The exception is Milwaukee, which had only 85 developed lots in its test window and gains from the larger pooled sample. Context improves the transferred model in all five cities. Portable features carry much of the signal, but local history still matters.

5.5 Manhattan rezonings

The three rezonings show the same contrast at the scale of a single decision (Table 8, Figure 3). Capacity projected 8,423 homes on development sites, and 5,207 when pro-rated to 2025. The likelihood-weighted projection gave 585; 561 had been built. The observed count lies within the model's 95% range in two of three rezonings. SoHo/NoHo is below it, with 3 homes built by 2025. Capacity describes full build-out by each review's build year, so the comparison concerns the near term, the horizon for which the likelihood model is trained.

Table 7: Transfer: AUC of a model fitted on the other cities' training windows / the city's own model, on the city's test window.

City	E	E+C	E+C+Z	E+C+Z+P
Los Angeles	0.719 / 0.864	0.817 / 0.892	0.828 / 0.894	0.837 / 0.894
Dallas	0.844 / 0.898	0.877 / 0.918	0.878 / 0.922	0.877 / 0.923
New York	0.825 / 0.892	0.874 / 0.914	0.868 / 0.919	0.866 / 0.919
Milwaukee	0.959 / 0.911	0.965 / 0.896	0.966 / 0.924	0.967 / 0.924
Chicago	0.742 / 0.824	0.808 / 0.888	0.804 / 0.891	0.822 / 0.891

Cities whose outcome is a building or a permit. A missing plan layer is coded as unknown.

Table 8: Three Manhattan rezonings: homes on development sites by 2025, observed and projected.

Rezoning (adopted)	Sites	Built by 2025	Capacity	Pro-rated to 2025	Likelihood-weighted (95%)
East Harlem (2017)	127	288	4,117	3,294	234 (79–489)
Inwood (2018)	47	270	2,859	1,334	213 (0–652)
SoHo/NoHo (2021)	77	3	1,447	579	138 (8–304)
All three		561	8,423	5,207	585

Capacity is full build-out by the review's build year; pro-rated scales it linearly to 2025. The likelihood comes from a redevelopment model trained only on outcomes known by the adoption year.

6 Discussion

What the layers are for. Across seven cities, zoning and the adopted plan add much less than existing conditions and neighbourhood context to predicting *where* development happens. They add more to predicting *what* it becomes, and they are needed to bound *how much* it could hold. The pattern is consistent with the causal literature. Zoning sets a ceiling and a direction, but whether a lot develops within a few years depends on its current use, value and surroundings (Rosenthal and Helsley, 1994; Weber et al., 2006; Charles, 2013). Where upzoning did not change construction in the short run (Freemark, 2020), little predictive value is to be expected. Where it did, the effect was strongest on vacant and underused parcels (Dong, 2024), which existing conditions already identify. Rezoning itself follows development opportunity (Gabbe, 2018), which helps explain why two of the regulatory gains vanished in the conservative variant. The plan results echo plan evaluation: plans mostly confirm existing use, so their predictive information lies where they depart from it. Their conformance with outcomes should not be read as influence (Berke et al., 2006; Feitelson et al., 2017).

Implications for planning support tools. Tools that combine existing, permitted and planned layers (Klosterman, 1999; Geertman and Stillwell, 2020) should not assume that each layer improves a projection. The results suggest three practices:

1. test each layer on the city's own history before relying on it, since its value varies by city and by question;
2. report likelihood-weighted projections beside capacity, which overstates near-term housing by orders of magnitude;
3. report a trend baseline, which here ranked districts better than any model-based projection.

These practices narrow the gap between claimed and demonstrated contributions that reviews of urban digital twins describe (Lei et al., 2023; Azadi et al., 2025).

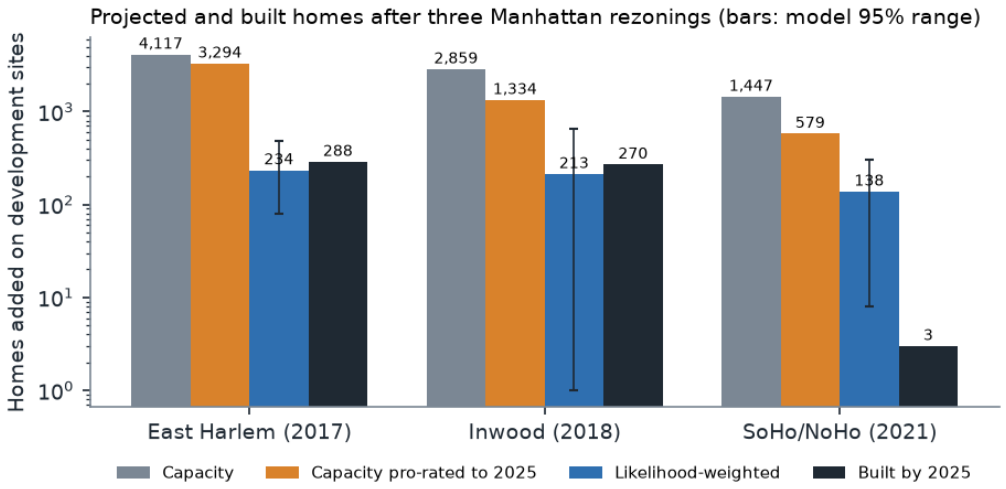


Figure 3: Homes on development sites after three Manhattan rezonings: capacity, capacity pro-rated to 2025, likelihood-weighted (with the model's 95% range) and built by 2025 (log scale).

Reusable data and methods. The crosswalks put ten cities' official codes on one schema as explicit tables, and tests check that every source code is mapped. The digitisation shows that plans published as documents need not stay outside quantitative analysis. Where the PDF is a vector export, a parcel layer can be recovered with errors of a few feet, and the shifted-map control provides a falsification test that does not require the true plan. The method relies only on a legend, filled shapes and a parcel layer, so it is not specific to Philadelphia. It failed where a map also draws neighbouring districts.

Limitations. The test covers 3,346,335 lots and 13,872 developments in seven large cities with different regulatory systems. That is enough to show that the value of zoning and plans varies and is often small, but not to estimate a general effect size. The main limitations are:

- the windows are short (one or two years) and, like the outcome definitions, differ by city;
- plan and zoning are current layers in Dallas and in parts of Phoenix, San Diego and Chicago; the conservative variant controls for this where dates exist, and in Dallas the gains are upper bounds;
- Chicago's outcome is a permit, not a completed building;
- context in Phoenix and San Diego lacks recent-change information in training;
- the crosswalks were written by hand, and some existing-use layers are a few years older than the windows;
- one model family is used, and AUC on rare events is complemented by average precision in the Appendix;
- the homes comparison rests on four cities and the rezoning replay on three cases;
- nothing here estimates what zoning or plans cause.

7 Conclusion

Zoning and adopted plans are the core of land-use regulation, and planning tools combine them with existing conditions to project development. Tested on observed history in seven US cities, the two layers add little to predicting where development happens once existing conditions and neighbourhood context are known. They help predict what development becomes, and they bound how much a site could hold. Projecting at capacity overstated near-term housing 76- to 193-fold. Weighting capacity by an out-of-time likelihood gave 0.81–2.64 times what was built, while a simple trend ranked districts better. The parcel layers for ten cities, the method for digitising plans published as maps and the test protocol are open, so any city's layers can be tested the same way before a tool relies on them.

Data and code availability

The code, the result files behind every number, the figures and a manifest of every input (URL, retrieval date and SHA-256) are in the PlanArena repository; `make reproduce` rebuilds the data and results from public sources. Each city's ingestion report documents coverage, crosswalks and estimated fields. Owner and mailing fields in assessor records were never requested or read.

Declaration of AI assistance

[To be confirmed by the author.] The author used Claude (Anthropic) to write and test code, run analyses and draft text. The author designed the study, reviewed the code and results, read the cited sources and takes responsibility for the content. Every reference was checked against the record its DOI or ISBN resolves to.

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A Additional results

Table 9: Out-of-time average precision and precision among the top 1% of lots.

City	Base rate	AP, E	AP, all	All blocks	Top 1%, E	Top 1%, all
Los Angeles	0.33%	0.139	0.147	E+C+Z+P	11.2%	12.8%
Dallas	1.34%	0.115	0.237	E+C+Z+P	12.6%	29.6%
New York	0.47%	0.073	0.122	E+C+Z	11.1%	15.1%
Milwaukee	0.06%	0.024	0.031	E+C+Z	2.3%	2.8%
Phoenix	0.36%	0.091	0.054	E+C+Z+P	9.7%	8.4%
San Diego	0.07%	0.002	0.002	E+C+Z+P	0.7%	0.6%
Chicago	0.26%	0.013	0.043	E+C+Z	2.4%	6.1%